SEL RTAC

Industrial Control Systems · SCADA Hub

COURSE TOPICS

1. **RTAC Architecture:** SEL-3505/3530/3555 hardware variants, I/O module selection, communication ports, and physical commissioning steps.
2. **ACSELERATOR Architect:** Project database structure, tags, aliases, protocol point mapping, revision control, and first-live-data workflow.
3. **DNP3 Configuration:** Outstation and master roles, point database mapping, event class assignment, unsolicited reporting, and SCADA latency parameters.
4. **Modbus Gateway:** TCP server configuration, DNP3-to-Modbus point mapping, register layout, data type conversion, and translation failure modes.
5. **IEC 61850 Integration:** GOOSE trip signals, Sampled Values, MMS operator control, and internal tag database mapping for each service type.
6. **Local Logic:** IEC 61131-3 programs, scan cycle behavior, event-driven automation, and design principles for WAN-independent operation.
7. **Security and Access Control:** Role-based access, serial port security, firmware update procedures, and NERC CIP implications per configuration decision.
8. **Field Scenarios:** Three case studies — automatic tap changer response, feeder protection with downstream reclosers, and sub-200ms load shedding.

Certification Relevance

- **SEL Certified Engineer** — SEL offers application engineer certification through their training center; RTAC configuration is a core track.
- **GICSP (GIAC)** — Substation automation, IEC 61850 security, and protocol gateway configuration appear in ICS security exam domains.
- **ISA CCST** — Protocol gateway integration and field device configuration tested at CCST Levels II and III.

Next Steps

1. Take DNP3 and IEC 61131-3 before RTAC. Protocol knowledge and programming skills are both prerequisites for field configuration work.
2. Download ACSELERATOR Architect (free from selinc.com) and configure a lab project against a virtual RTAC.
3. Attend SEL application engineering training in Pullman, WA or at a regional event. The in-person lab is the fastest path to real proficiency.
4. GICSP pairs well with RTAC expertise for engineers working in power utilities and substations.